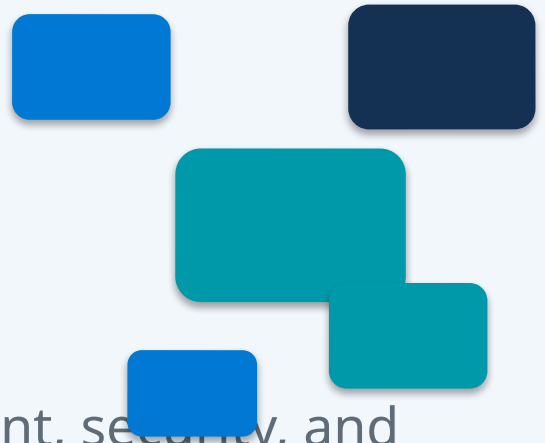
AZURE DATA FACTORY

Screenshots & Practical Guide



A visual walkthrough of setup, authoring, monitoring, management, security, and linked services.

29 annotated screenshots • Key concepts • Practical use cases

How to use this guide

Each section pairs the original Azure portal screenshot with a short explanation and a real-world use case.

01 **Create & configure**

Basics, Git, networking, encryption, tags

02 **Author & build**

Workspace, pipelines, datasets, data flows, Power Query

03 **Monitor**

Pipeline runs, trigger runs, runtimes, debug, alerts

04 **Manage**

Linked services, settings, Purview, Git, ARM templates

05 **Security & reuse**

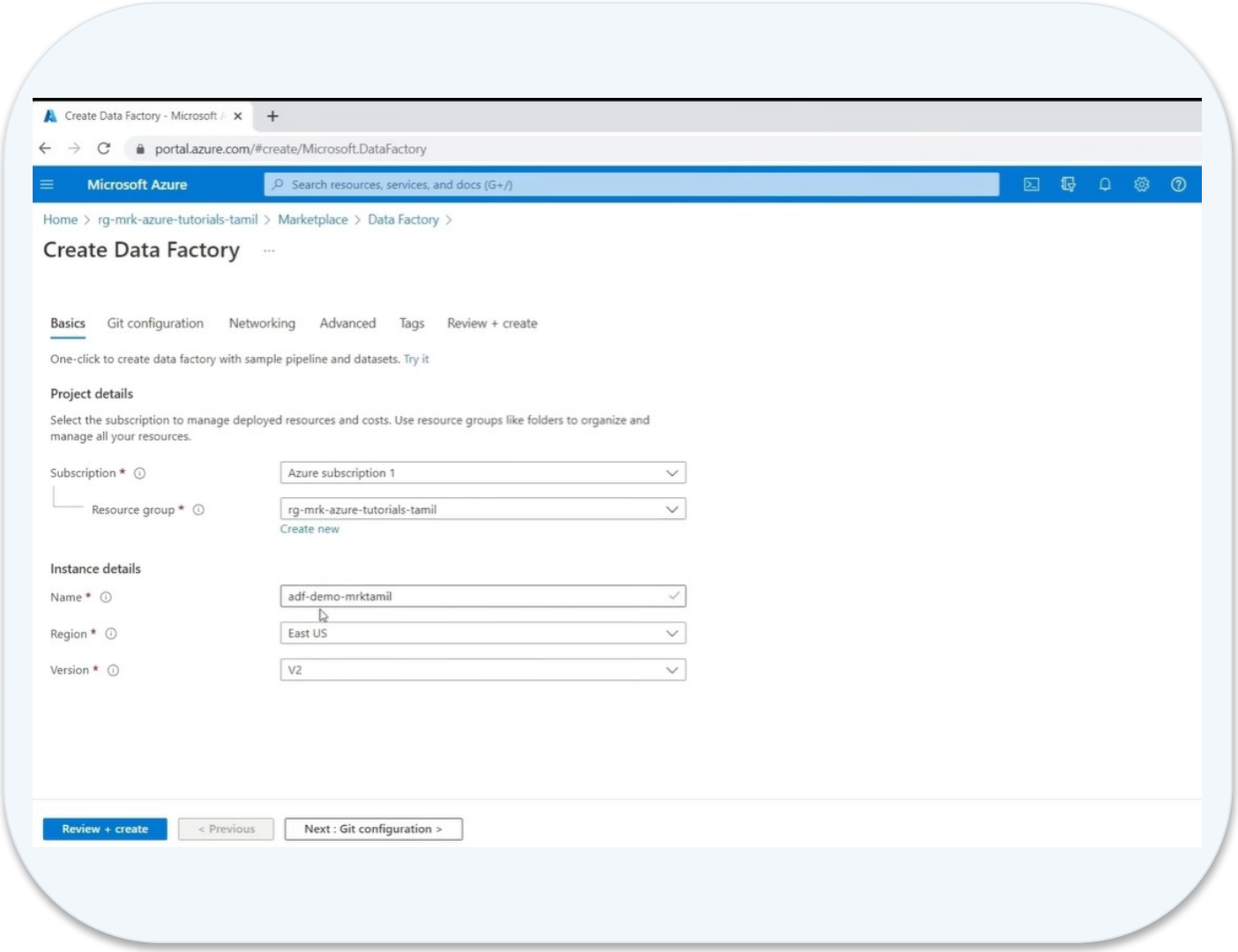
Triggers, parameters, libraries, credentials, keys

06 **Connect data**

Integration runtime setup and ADLS Gen2 linked service

Tip: “0 items” or “No results” usually means no matching runs/resources are listed—not automatically that the service is broken.

Create Data Factory – Basics



PURPOSE

Create the Azure Data Factory resource that will host your data integration work.

KEY POINTS

Subscription: Azure account that manages resources and billing.

Resource group: Container for related Azure resources.

Name: Unique name for the Data Factory.

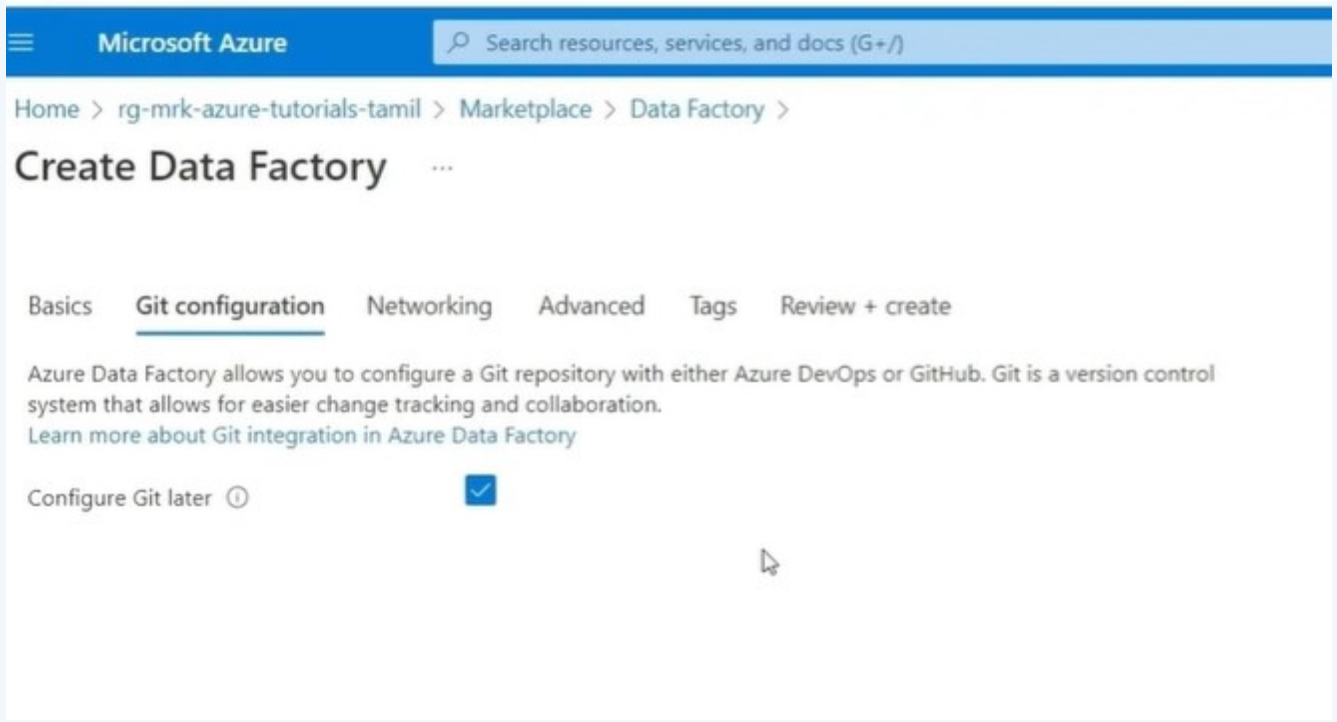
Region: Azure location where the resource is deployed.

Version: V2 is the Data Factory generation shown.

USE CASE

Create a factory as the starting point for pipelines that move sales data from databases and files into a central data lake.

Git Configuration During Creation



PURPOSE

Choose whether to configure source control while creating the factory.

KEY POINTS

Azure DevOps or GitHub can store pipeline definitions and track changes.

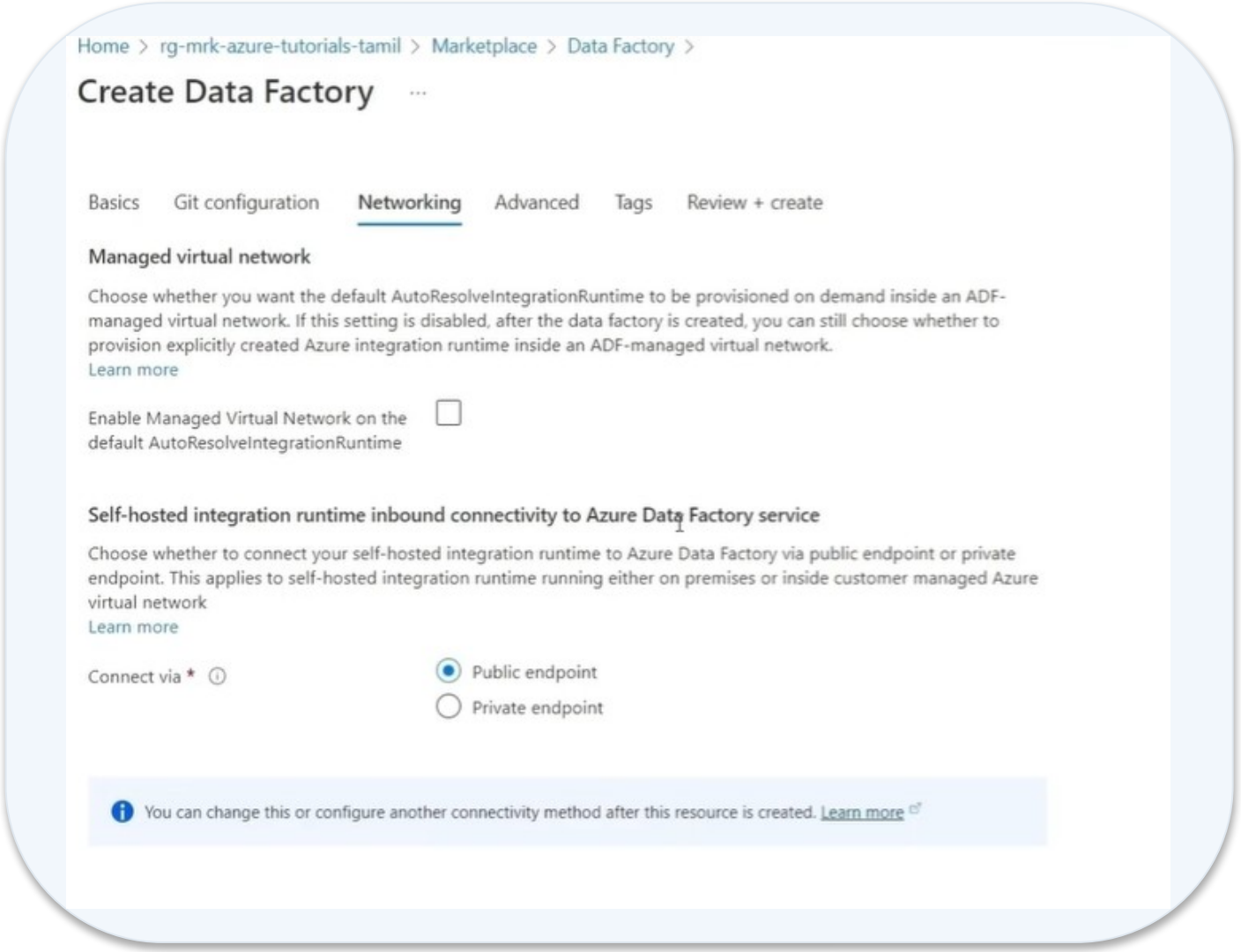
Configure Git later lets you skip repository setup during creation.

Git is useful for collaboration, review, and version history.

USE CASE

A developer commits a pipeline change to Git so teammates can review it before publishing.

Networking



PURPOSE

Control how Azure Data Factory connects to services and data sources.

KEY POINTS

Managed Virtual Network can provide network isolation for supported connections.

Public endpoint: connect through a public service endpoint.

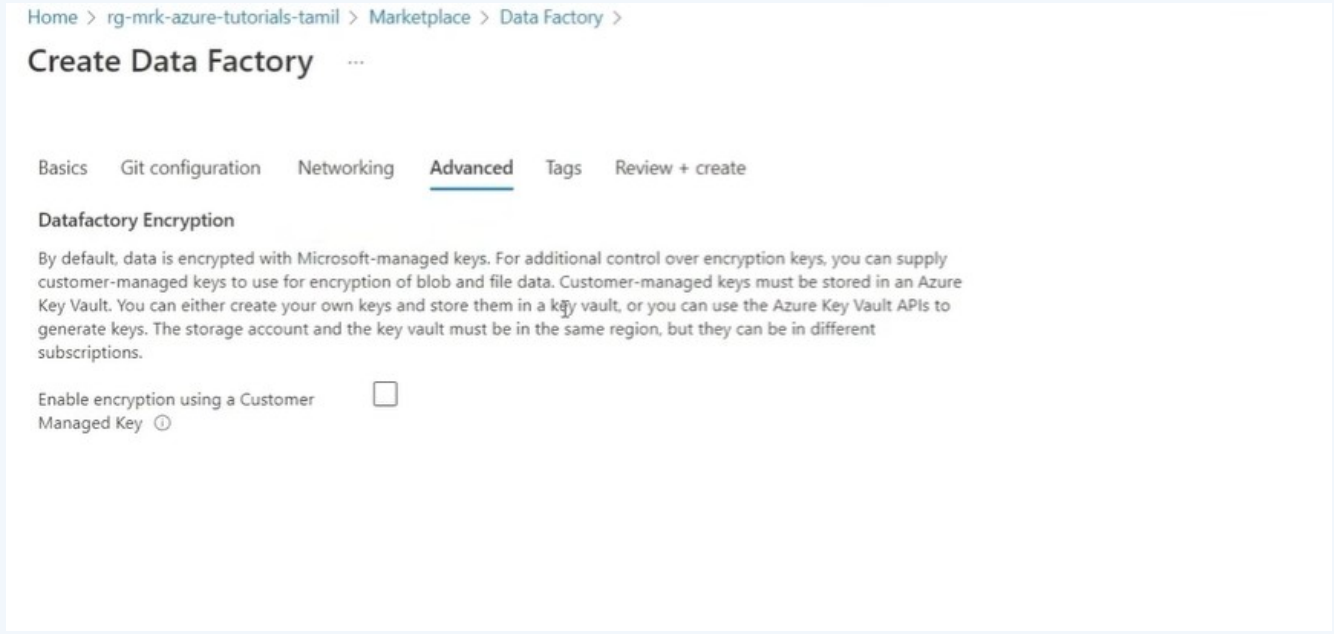
Private endpoint: use private IP connectivity where supported.

Choose connectivity based on security and network requirements.

USE CASE

A company processing sensitive data may use private connectivity to reduce exposure to the public internet.

Advanced – Encryption



PURPOSE

Choose encryption-key management for the Data Factory.

KEY POINTS

Microsoft-managed keys are the default and are managed by Azure.

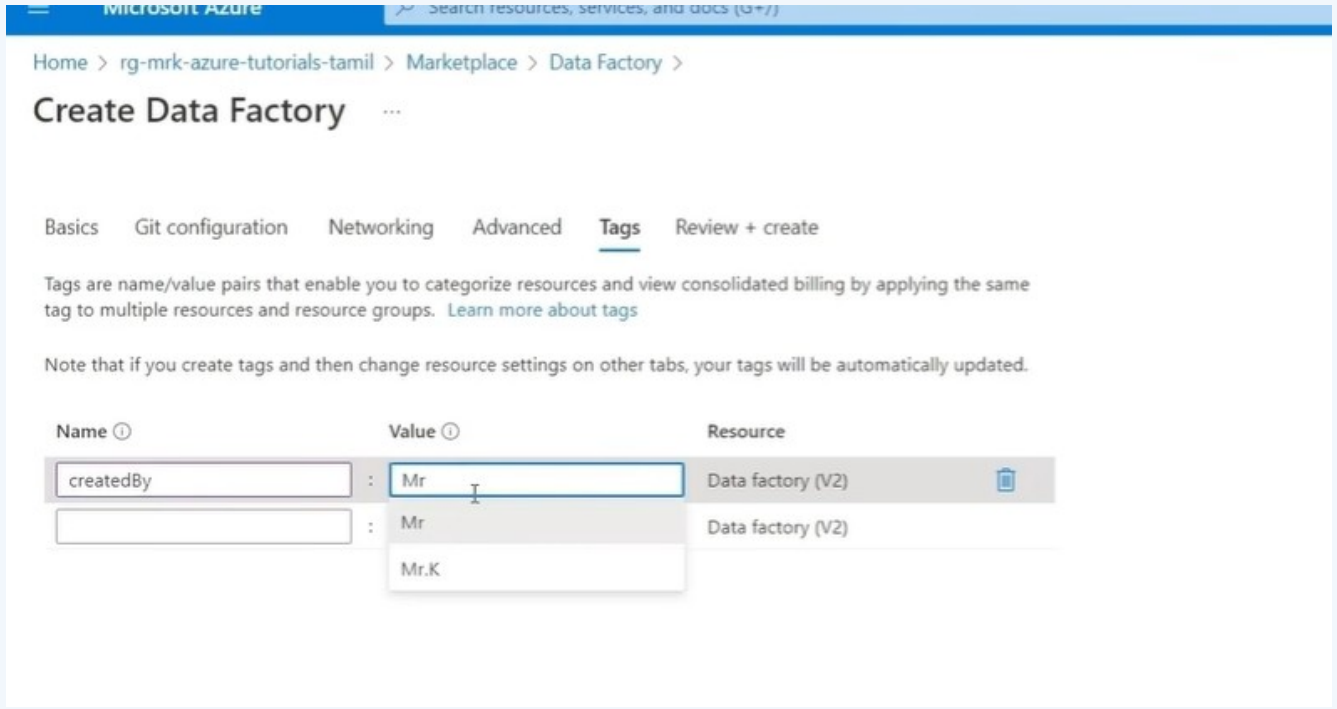
Customer-managed keys (CMK) can be configured through Azure Key Vault.

CMK offers more control over key access and rotation but requires extra setup.

USE CASE

A regulated organization manages encryption keys to align with its security and compliance policies.

Tags



PURPOSE

Tags are name/value pairs that help organize Azure resources and report costs.

KEY POINTS

Example: createdBy = Mr.K

Example: environment = Development

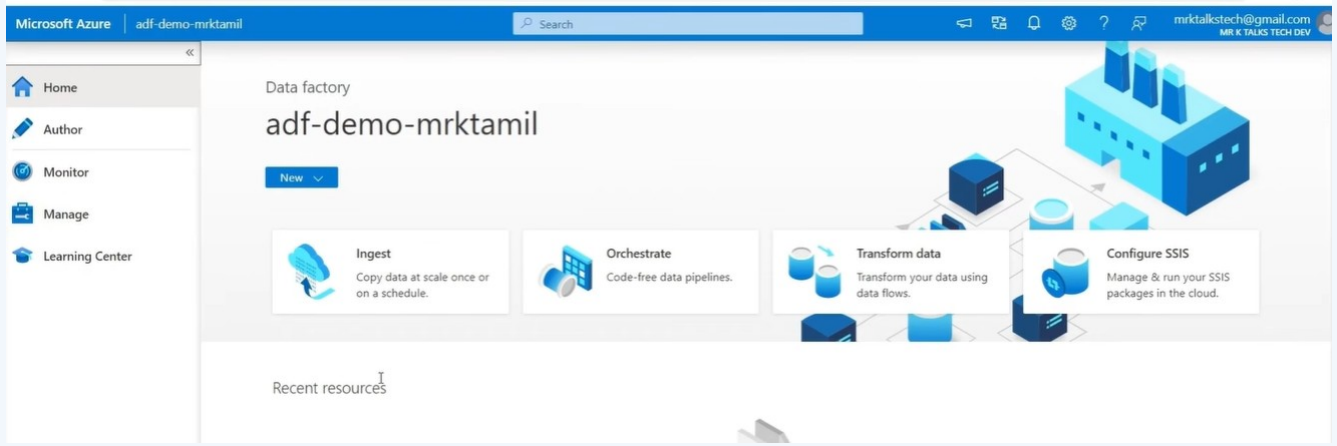
Example: department = DataEngineering

Use consistent tags to identify ownership and group resources.

USE CASE

A team reviews resource ownership and spending by department or environment.

Data Factory Workspace Overview



PURPOSE

The main workspace is where you author, manage, and monitor data integration.

KEY POINTS

Author: create pipelines, datasets, linked services, and data flows.

Monitor: inspect pipeline runs, failures, and duration.

Manage: configure connections, runtimes, triggers, and security.

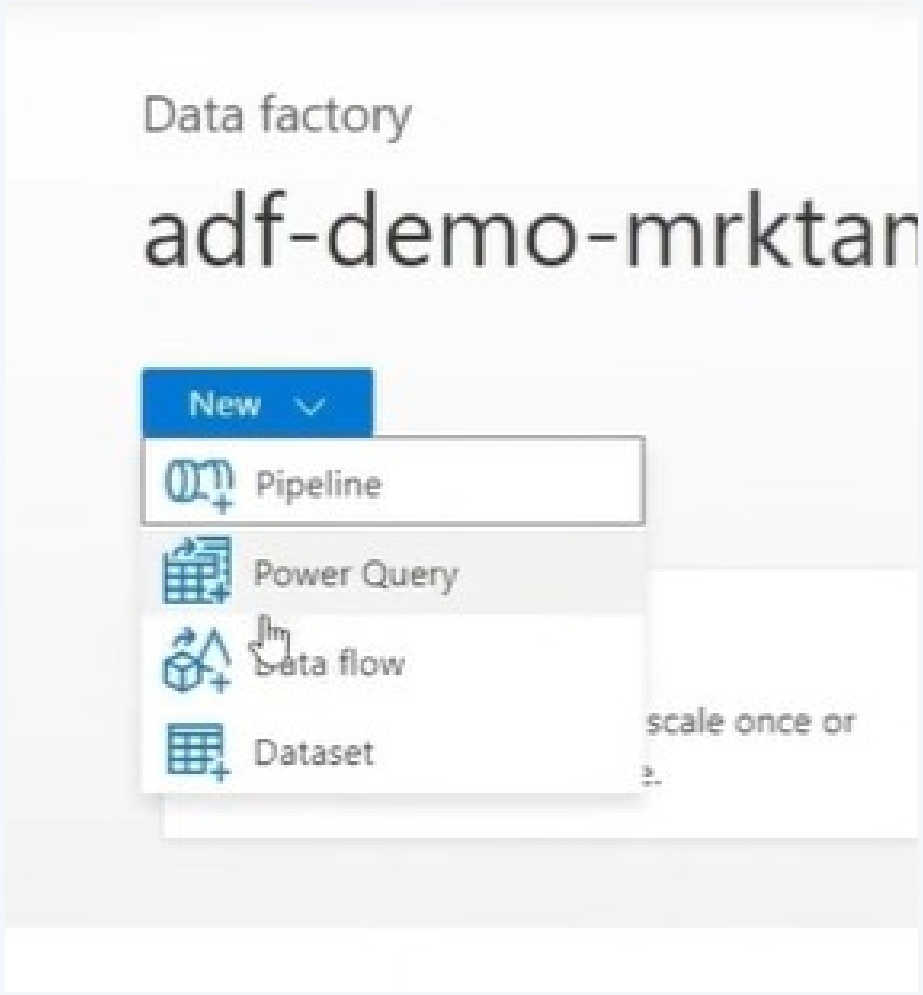
Learning Center: tutorials, videos, and templates.

Ingest, orchestrate, transform, and run SSIS packages from the workspace.

USE CASE

Automate sales-data ingestion, transform it, and load it for analytics and Power BI reporting.

New Menu – Create Components



PIPELINE

Orchestrates activities into a workflow.

KEY POINTS

Power Query: low-code data preparation.

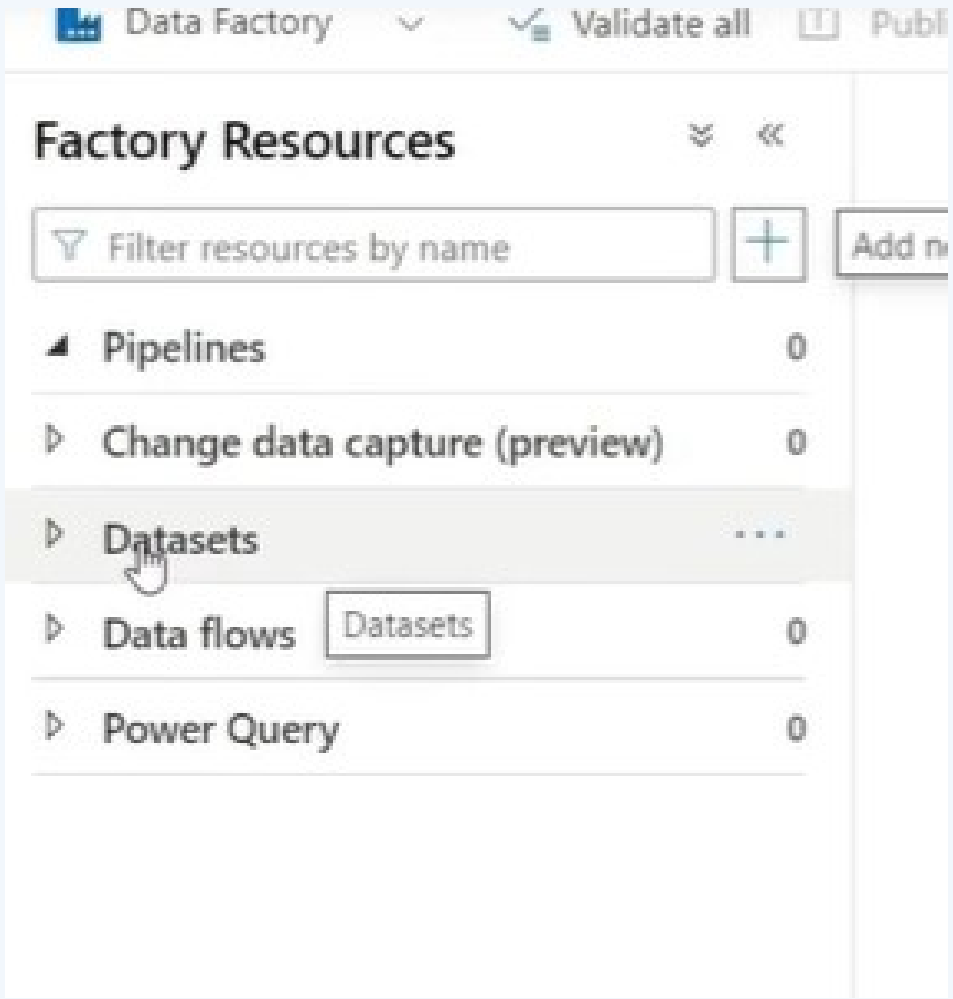
Data flow: visual transformations such as joins, filters, and aggregates.

Dataset: describes the location and format of data used by activities.

USE CASE

Build a pipeline that copies a CSV file, transforms records, and loads the results into a warehouse.

Factory Resources



PURPOSE

Factory Resources lists the main building blocks used to author data workflows.

KEY POINTS

Pipelines: coordinate and execute activities.

Datasets: describe source or destination data.

Data flows: visually transform data.

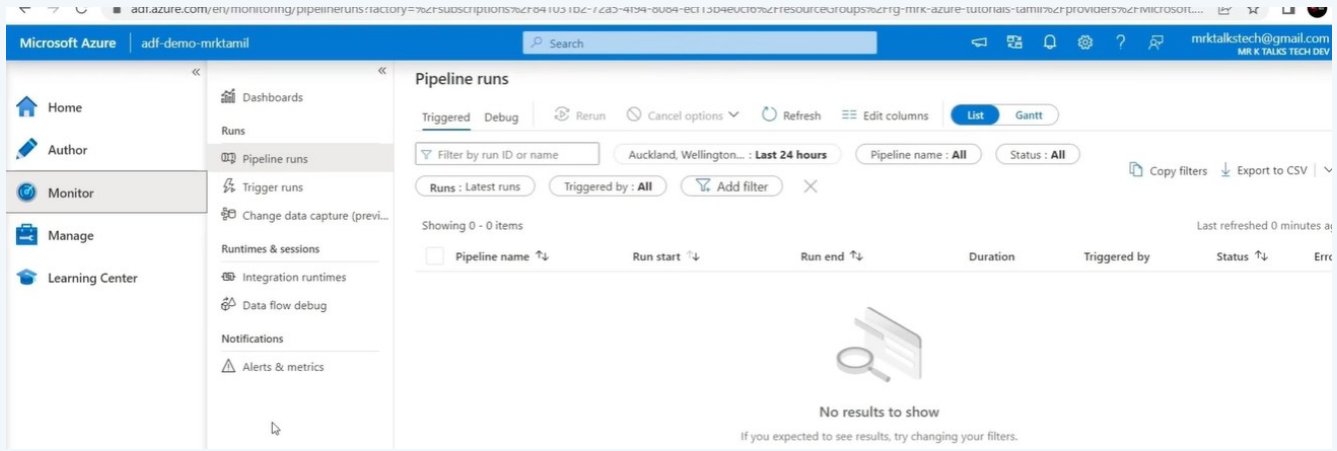
Power Query: prepare data with a low-code experience.

Change Data Capture (preview): support processing data changes where configured.

USE CASE

Create source/destination datasets, use a pipeline for orchestration, and add transformations when needed.

Monitor – Pipeline Runs



PURPOSE

View executions of pipelines and investigate their outcomes.

KEY POINTS

Check pipeline name, start/end time, duration, trigger, status, and errors.

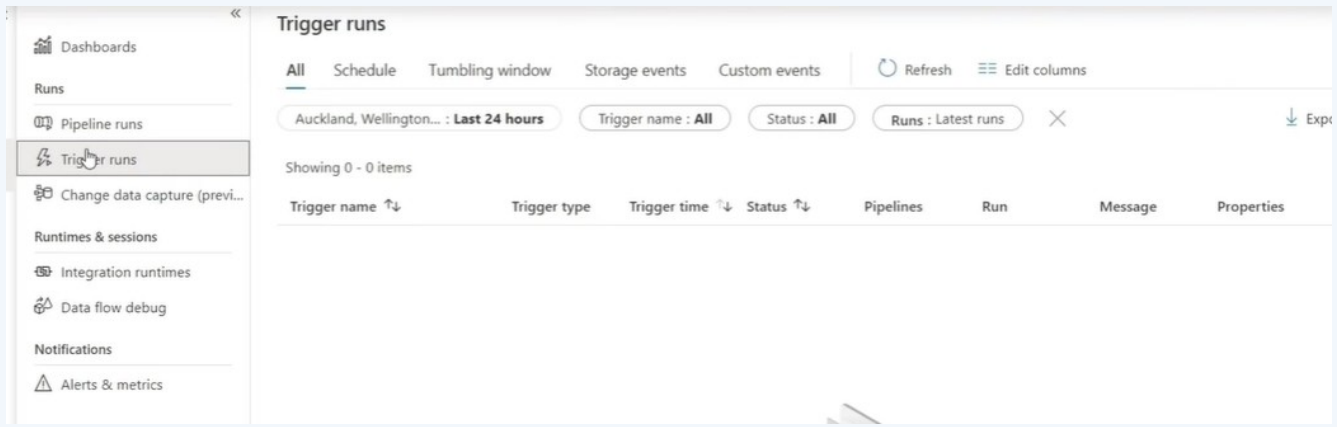
Use filters to find a run by name, status, or time range.

Rerun a pipeline after investigating and fixing a failure.

SCREENSHOT NOTE

The page shows 0 items. No runs match the current filters/time range; it does not necessarily mean there is an error.

Monitor – Trigger Runs



PURPOSE

Inspect executions of triggers that start pipelines automatically.

KEY POINTS

Schedule: start at a specified time or recurring interval.

Tumbling window: process fixed, continuous time windows.

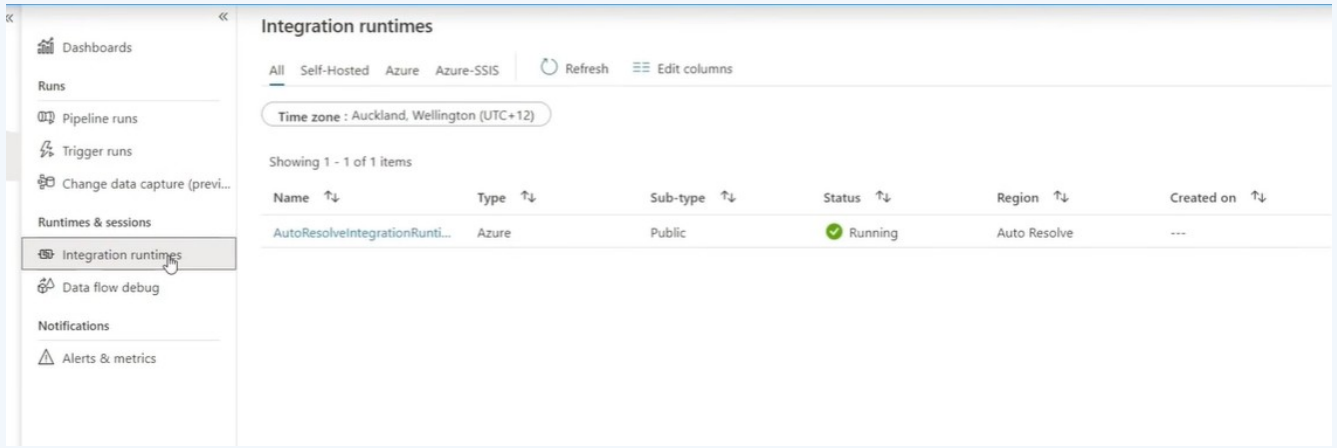
Storage events: start when a storage event occurs.

Custom events: start when a matching event is received.

SCREENSHOT NOTE

0 items are shown for the selected time range and filters. Create and activate a trigger before expecting runs.

Monitor – Integration Runtimes



PURPOSE

Check the integration runtime available to execute supported data integration activities.

KEY POINTS

The screenshot shows AutoResolveIntegrationRuntime.

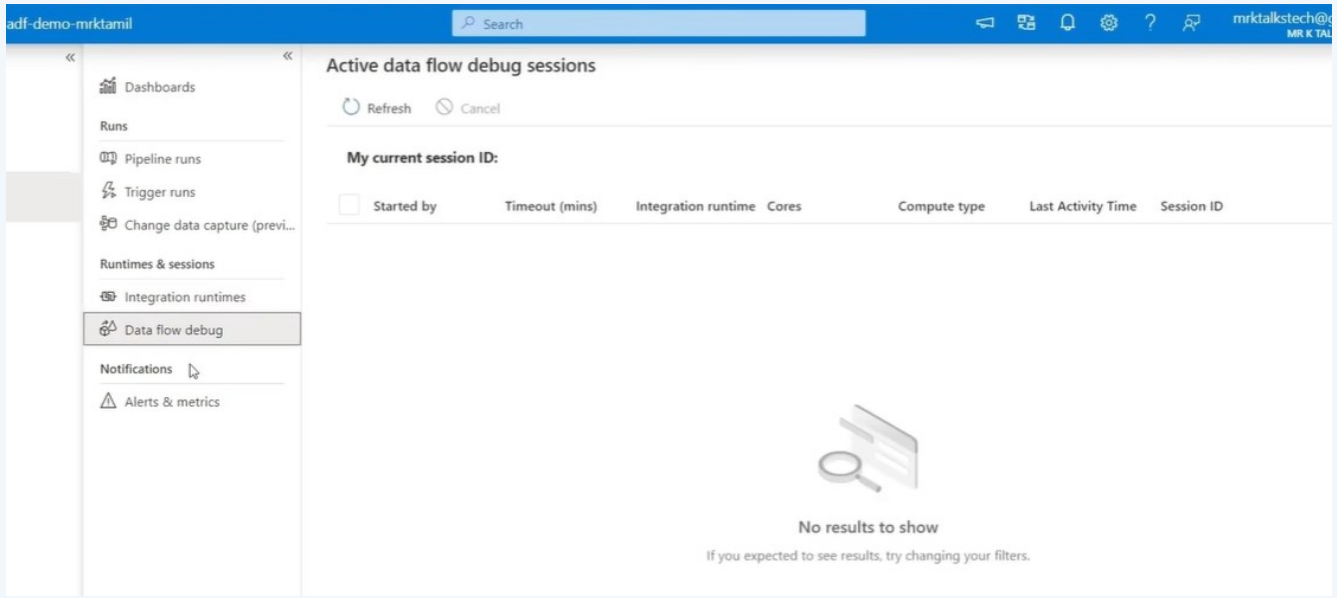
Type: Azure; subtype: Public; status: Running; region: Auto Resolve.

The Azure runtime provides compute for supported data movement and activities.

USE CASE

Copy data from Azure SQL Database to Azure Data Lake Storage using an Azure Integration Runtime.

Monitor – Data Flow Debug



PURPOSE

Test and troubleshoot mapping data flows interactively before production execution.

KEY POINTS

Debug sessions use compute from an integration runtime.

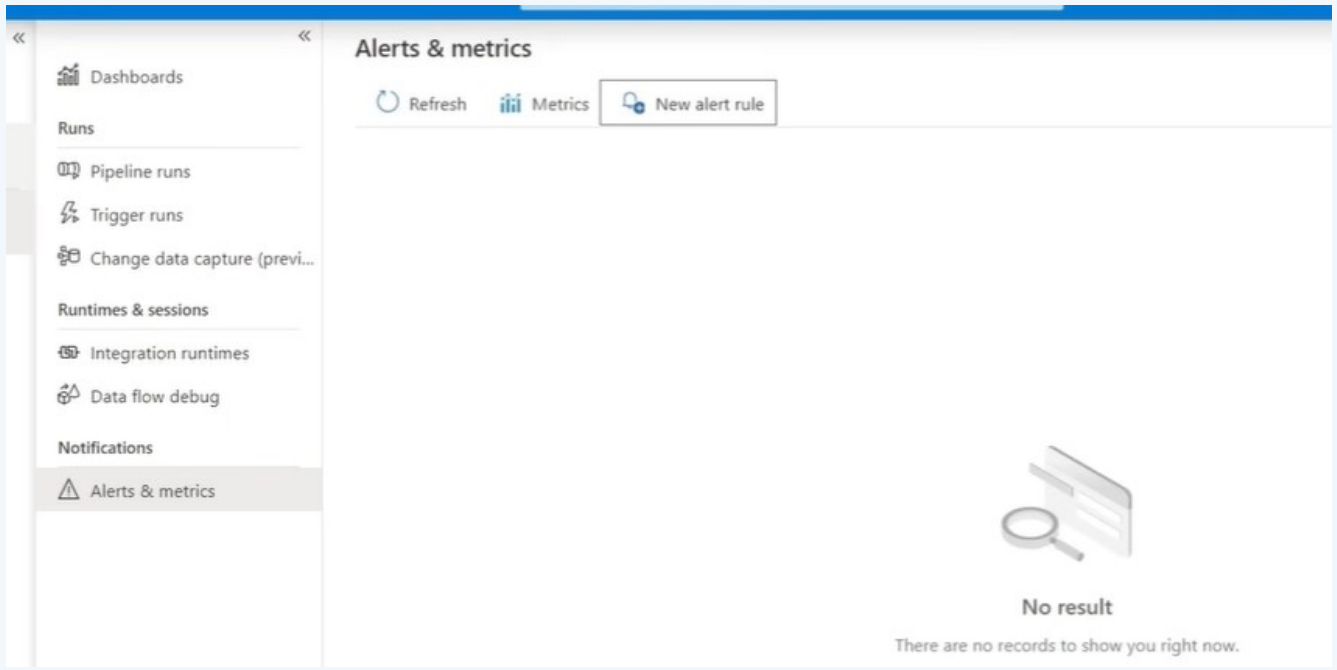
Session details can include timeout, cores, compute type, and last activity.

Refresh the list or cancel a selected active session.

SCREENSHOT NOTE

No active debug sessions are listed in the screenshot.

Monitor – Alerts & Metrics



PURPOSE

Review supported monitoring metrics and configure alerts for selected conditions.

KEY POINTS

Metrics: inspect available monitoring measurements.

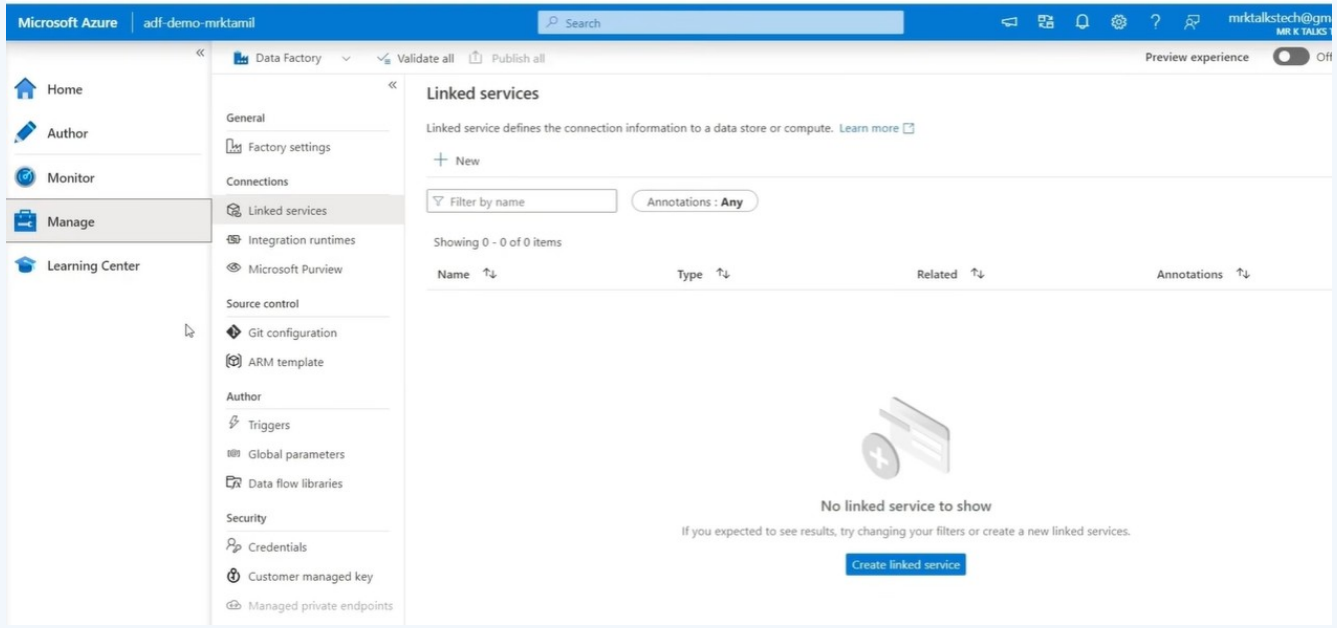
New alert rule: define a condition and configure notifications.

Alerts can help teams notice failures or unusual behavior quickly.

USE CASE

Create an alert for a supported pipeline-monitoring condition so the support team can investigate promptly.

Manage – Linked Services (Empty)



PURPOSE

A linked service defines connection information for a data store or compute service.

KEY POINTS

Create a connection for each required source or destination.

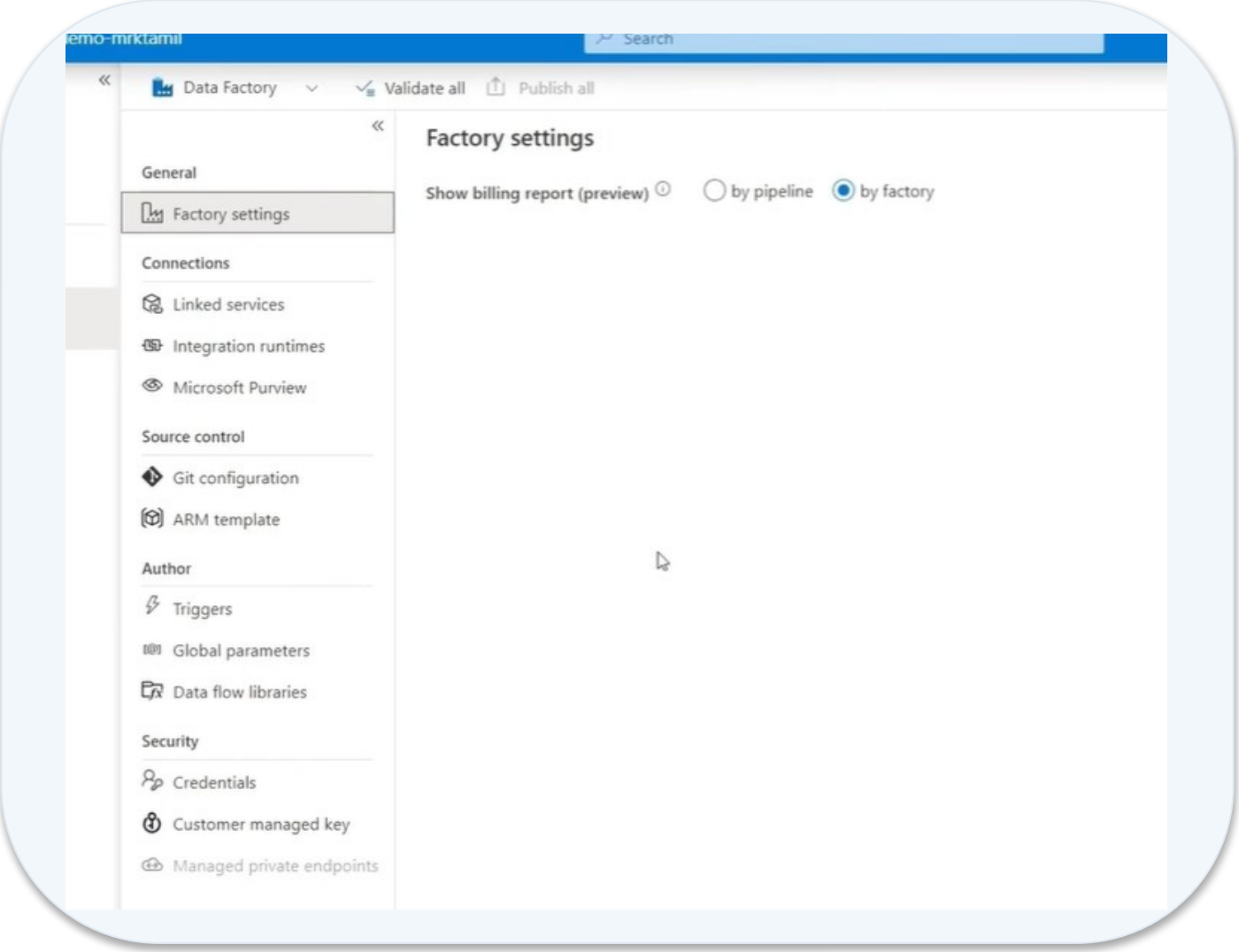
Examples include Azure SQL Database, Blob Storage, and Data Lake Storage.

Use the connection in pipeline activities or datasets.

SCREENSHOT NOTE

No linked services are configured yet. Create the required connections before using them in a pipeline.

Manage – Factory Settings



PURPOSE

Factory-level settings include the billing-report view shown in this screenshot.

KEY POINTS

By pipeline: view billing information grouped by pipeline where supported.

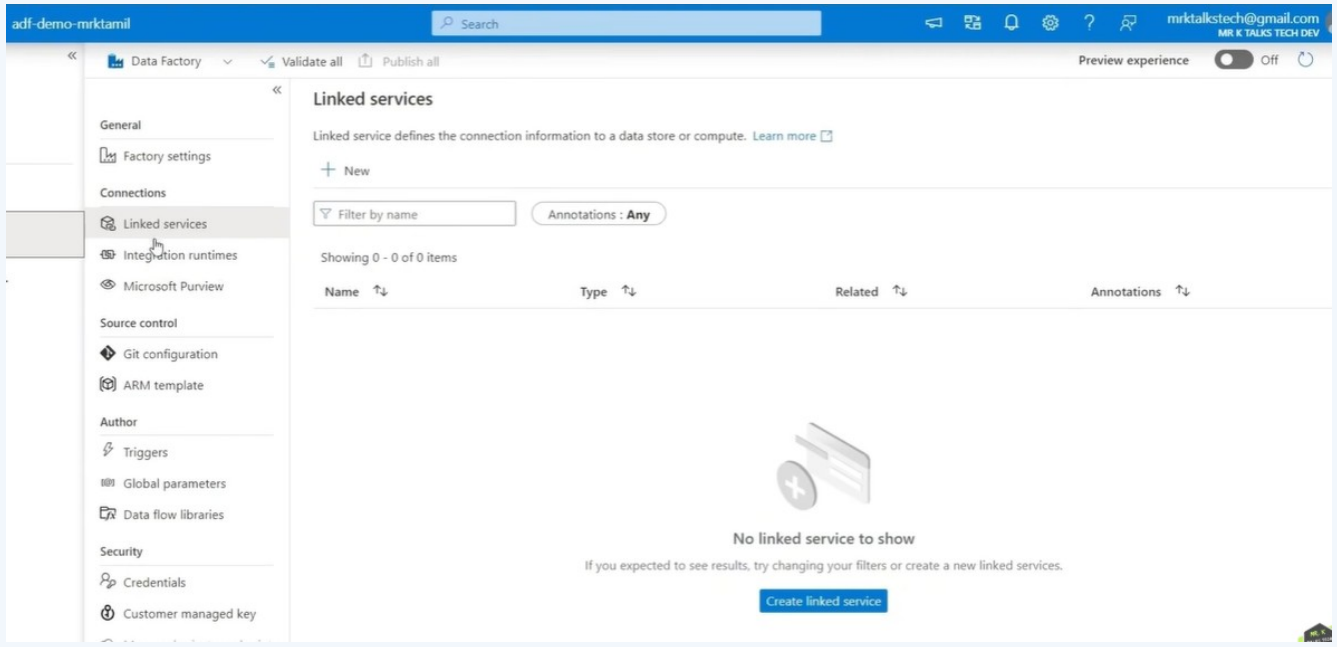
By factory: view billing information for the overall Data Factory.

Use the view that best supports your cost investigation.

USE CASE

Compare pipeline-level costs to identify which workflows contribute most to overall usage.

Manage – Linked Services



PURPOSE

Create and manage the connections used by your factory.

KEY POINTS

New: add a linked service.

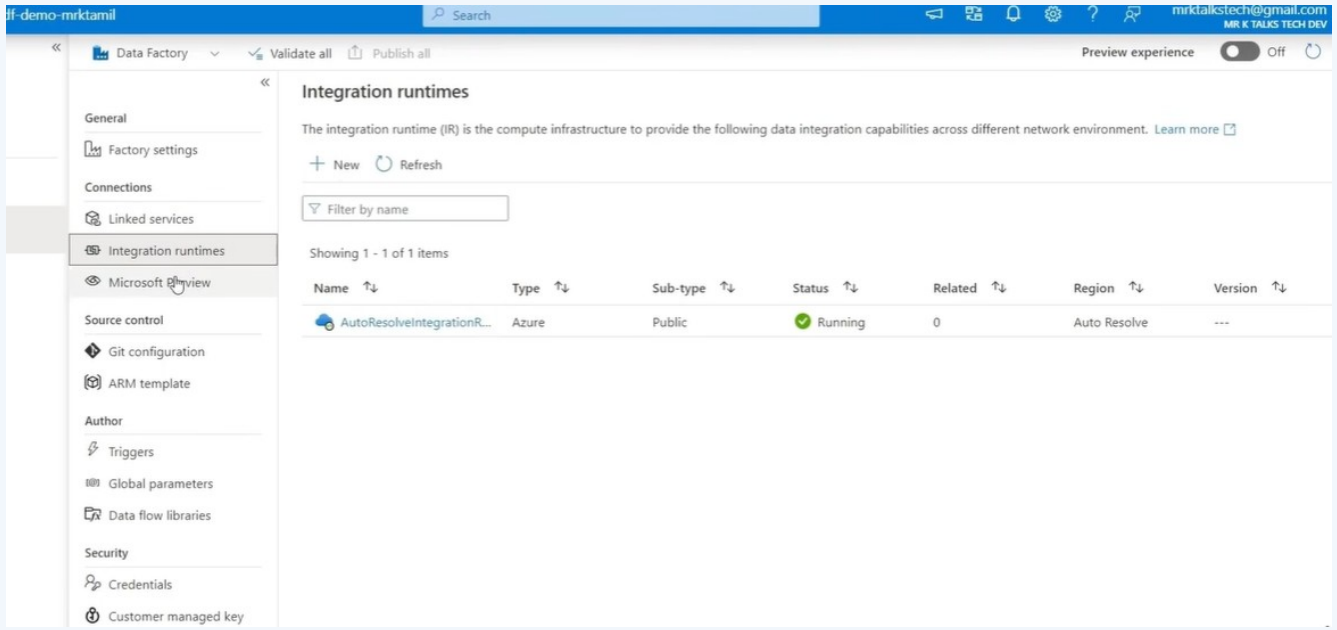
Filter by name: find an existing connection.

Related resources and annotations help organize connections.

USE CASE

Create one linked service for Azure SQL Database and another for Data Lake Storage, then use both in a Copy Data activity.

Manage – Integration Runtimes



PURPOSE

Configure the compute and connectivity infrastructure for data integration.

KEY POINTS

Azure: Azure-managed runtime for supported activities.

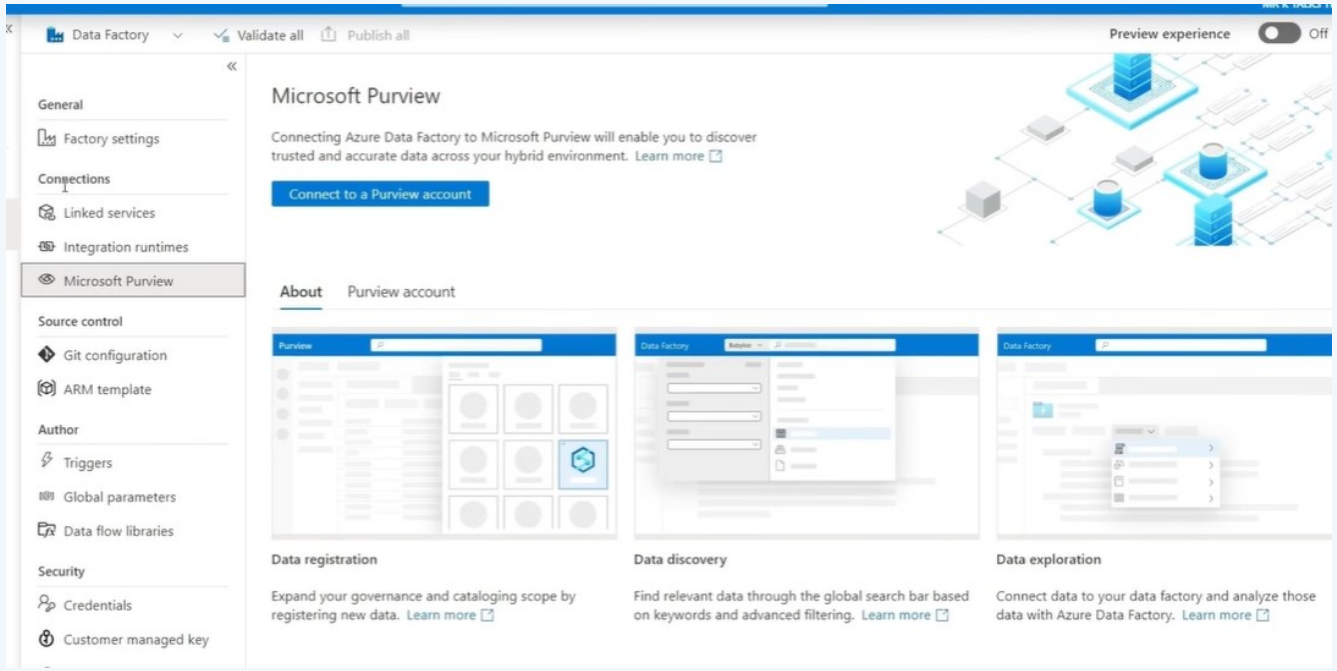
Self-hosted: access on-premises or private-network data sources.

Azure-SSIS: run supported SSIS packages in Azure.

SCREENSHOT NOTE

AutoResolveIntegrationRuntime is shown as Running, with Type Azure and Region Auto Resolve.

Manage – Microsoft Purview



PURPOSE

Connect Data Factory to Microsoft Purview for data discovery and governance capabilities.

KEY POINTS

Data registration: register data sources/assets.

Data discovery: search for relevant data assets.

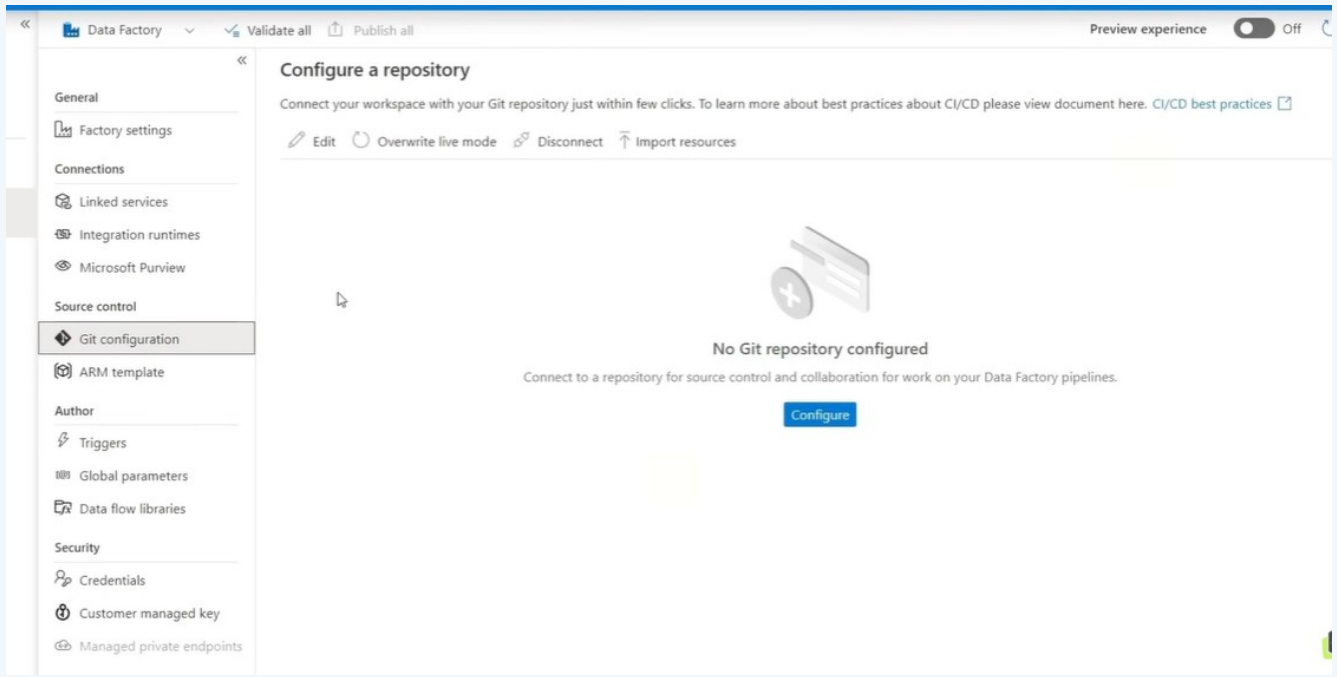
Data exploration: examine assets and relationships.

Lineage and governance depend on configured sources and supported integrations.

USE CASE

Find the origin of a customer dataset and understand related data assets across systems.

Manage – Git Configuration



PURPOSE

Connect the authoring workspace to a Git repository for source control.

KEY POINTS

Configure: connect to a repository.

Edit: change repository settings.

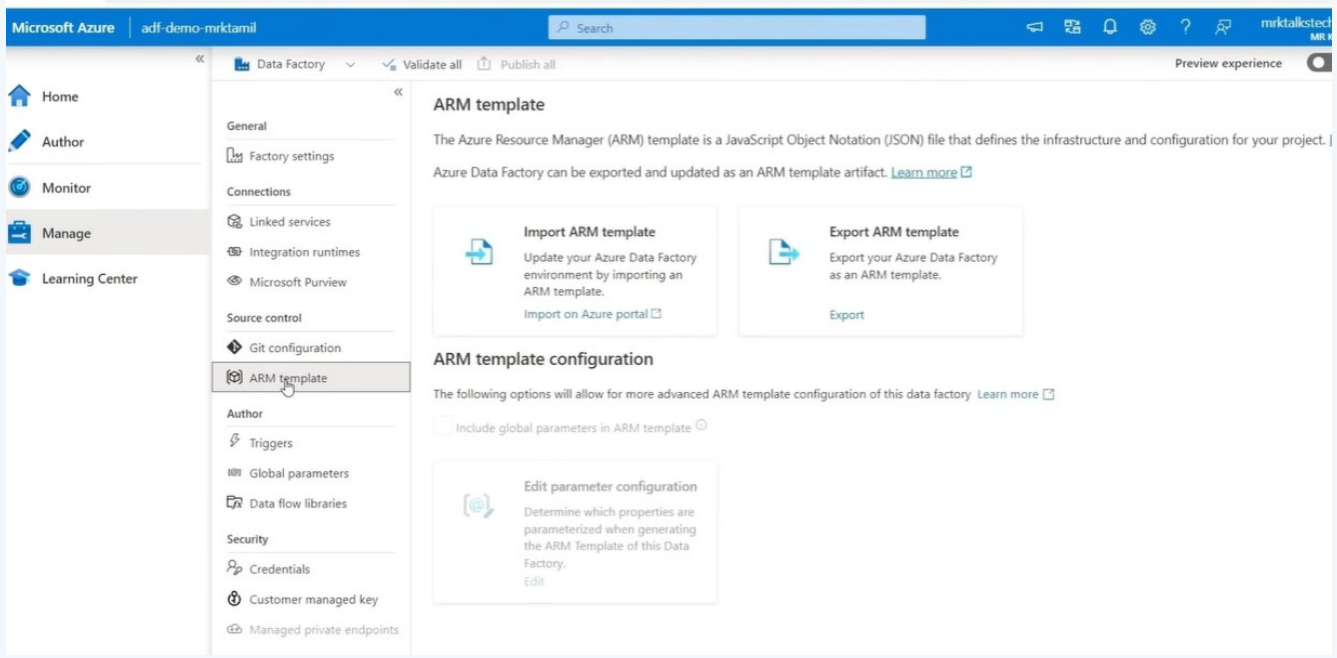
Overwrite live mode: replace live resources with content from the collaboration branch.

Disconnect: remove the connection; Import resources can bring resources into the workspace.

SCREENSHOT NOTE

No Git repository is configured in this screenshot.

Manage – ARM Template



PURPOSE

Export and deploy Data Factory configuration using Azure Resource Manager (ARM) templates.

KEY POINTS

Export ARM template: package supported factory resources as JSON template artifacts.

Import ARM template: apply a template to create or update resources.

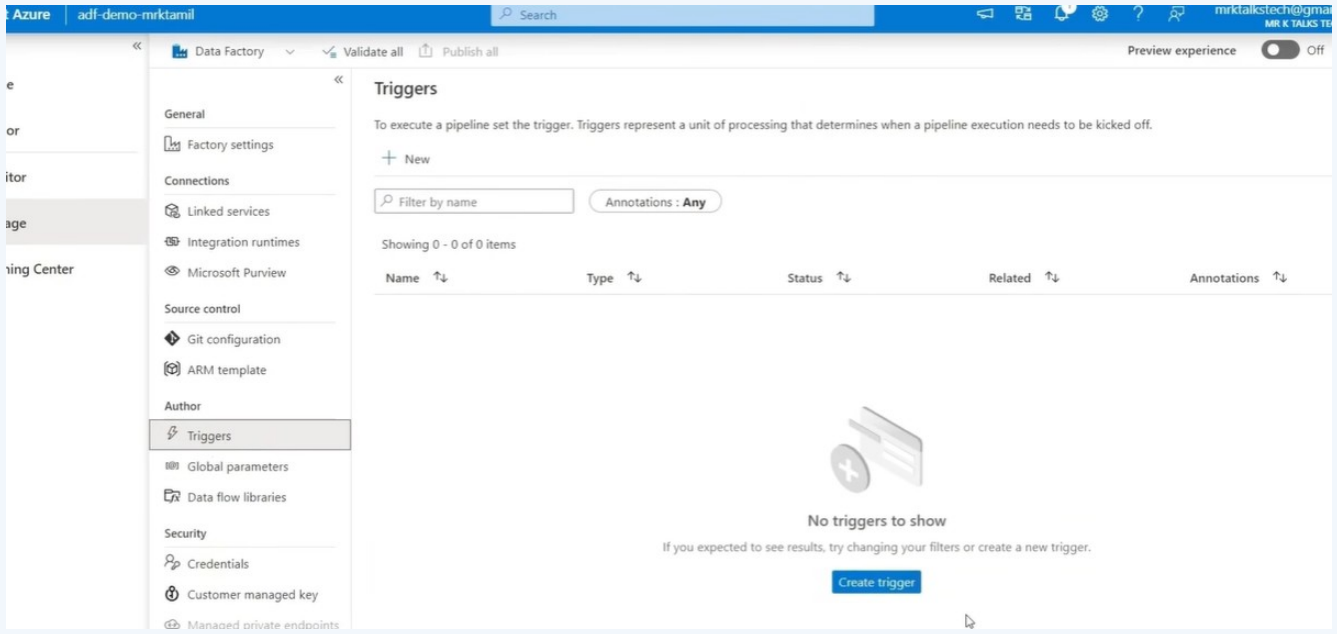
Parameter configuration controls which properties are parameterized.

Global-parameter template option is marked deprecated in the screenshot.

USE CASE

Promote factory configuration from Development to Test and Production with repeatable deployments.

Manage – Triggers



PURPOSE

Define when a pipeline should start.

KEY POINTS

Schedule trigger: run at a time or recurring interval.

Tumbling window: run over fixed time windows.

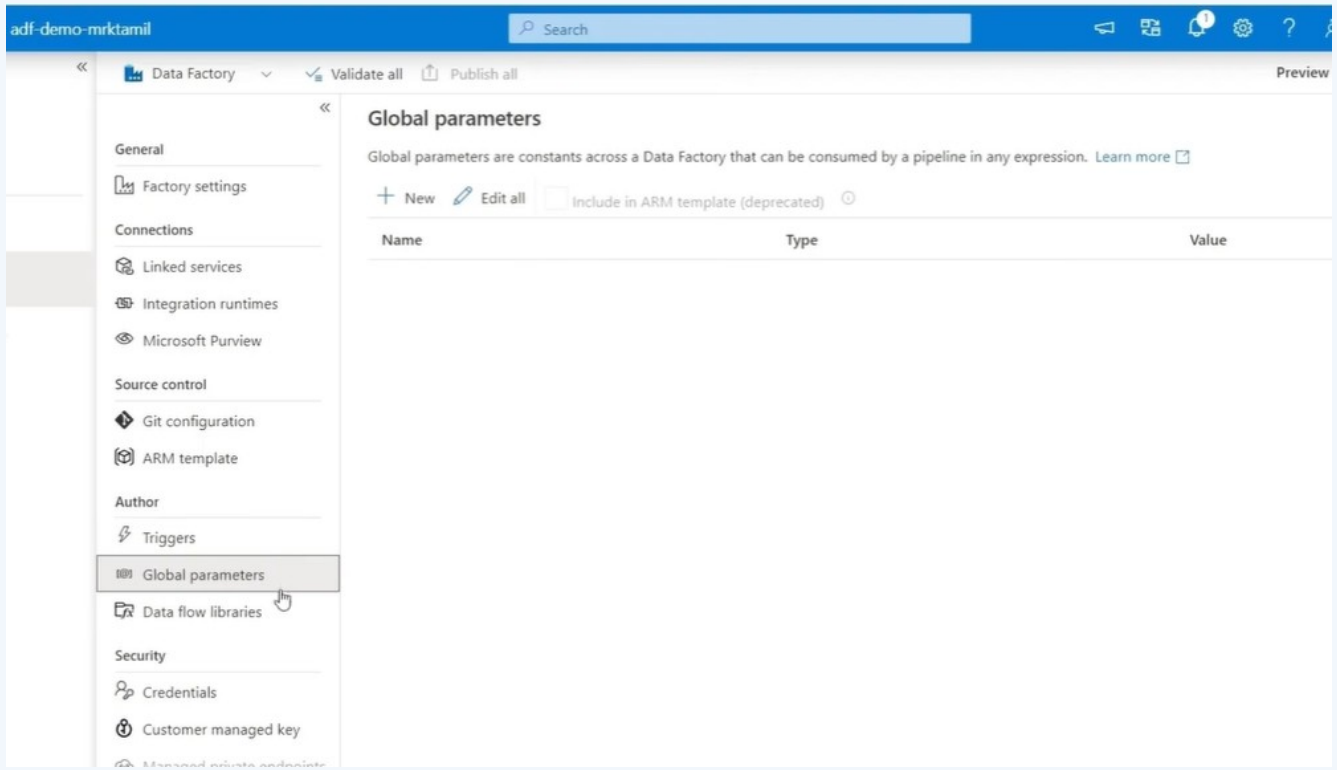
Storage event: start when a matching storage event occurs.

A trigger must be configured and activated to start pipeline runs.

SCREENSHOT NOTE

No triggers have been created in this factory yet.

Manage – Global Parameters



PURPOSE

Global parameters are reusable factory-level values available to pipeline expressions.

KEY POINTS

Create a parameter with a name, type, and value.

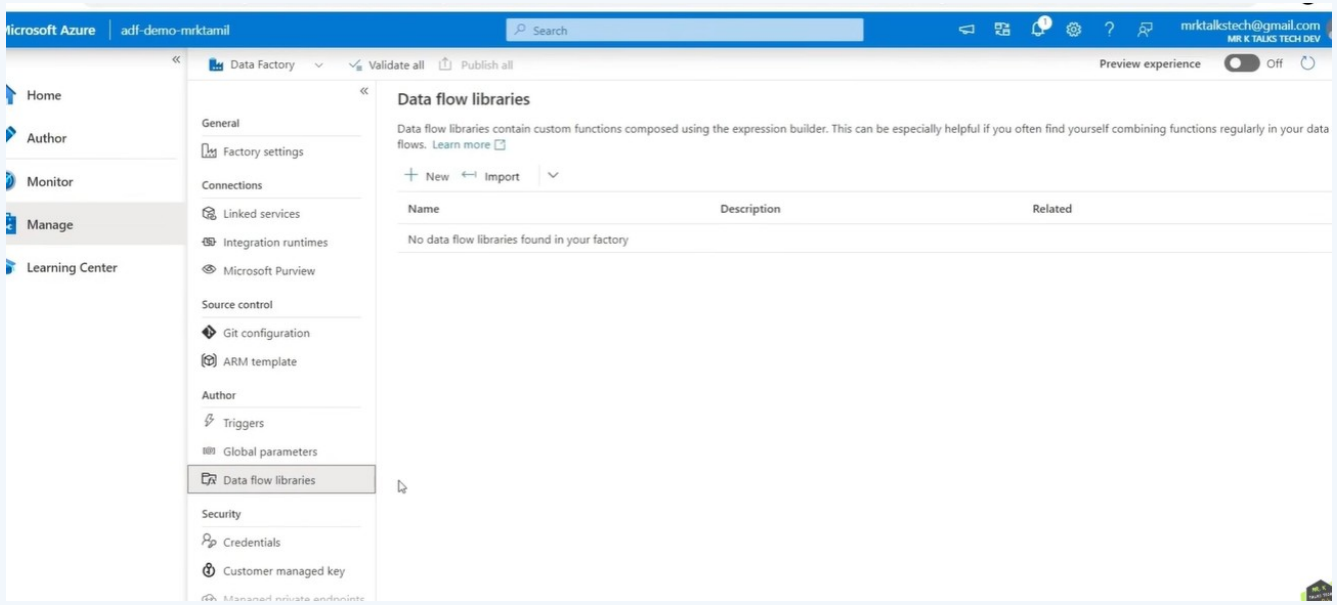
Example: storageFolder (String) = salesdata.

Reference the shared value from multiple pipelines instead of repeating literals.

USE CASE

Change a common folder value in one place instead of editing several pipelines.

Manage – Data Flow Libraries



PURPOSE

Data flow libraries contain reusable custom functions for mapping data flows.

KEY POINTS

New: create a library; Import: bring in an existing library.

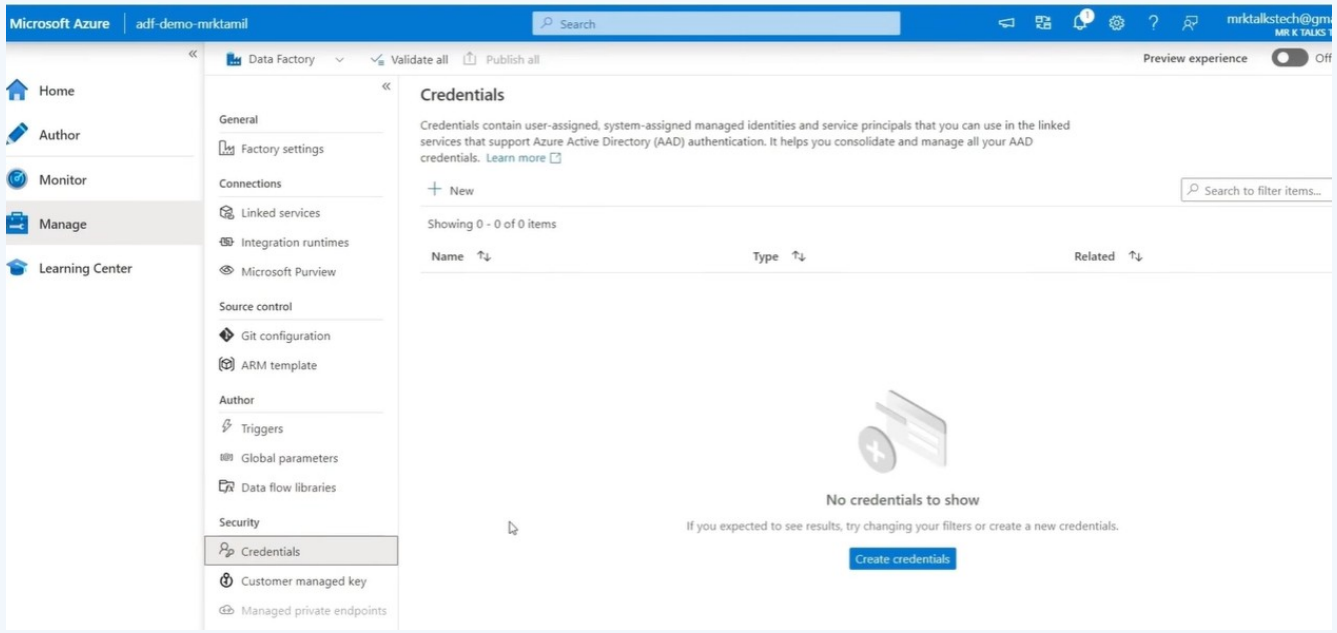
Reuse repeated transformation logic where supported.

This can reduce duplicated expressions and help standardize transformations.

USE CASE

Reuse a custom function that standardizes customer names across multiple data flows.

Manage – Credentials



PURPOSE

Credentials provide identity information for authentication in supported linked services.

KEY POINTS

Supported options can include managed identities and service principals.

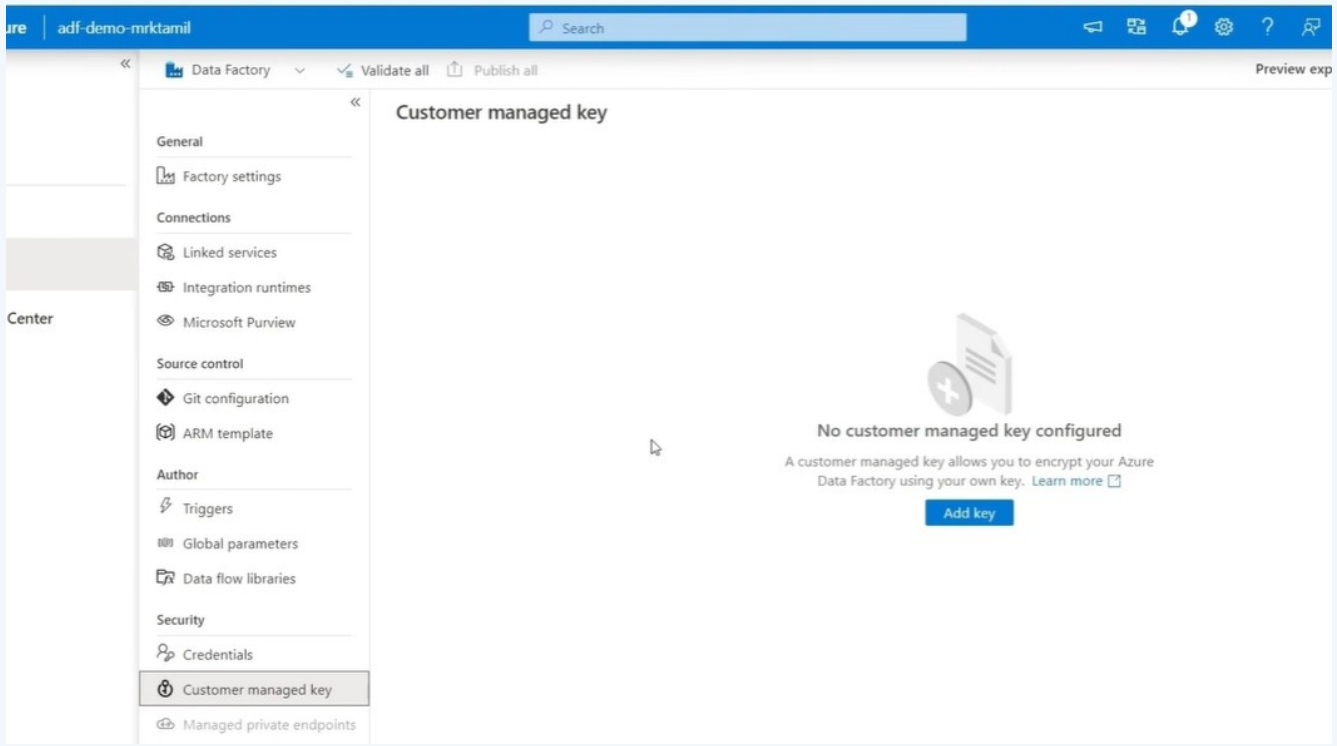
Grant only the permissions required by the target service.

Avoid embedding secrets directly in pipeline definitions.

SCREENSHOT NOTE

No credentials are listed in the screenshot.

Manage – Customer-Managed Key



PURPOSE

Configure a customer-managed encryption key for the Data Factory where supported.

KEY POINTS

Keys are typically stored and managed through Azure Key Vault.

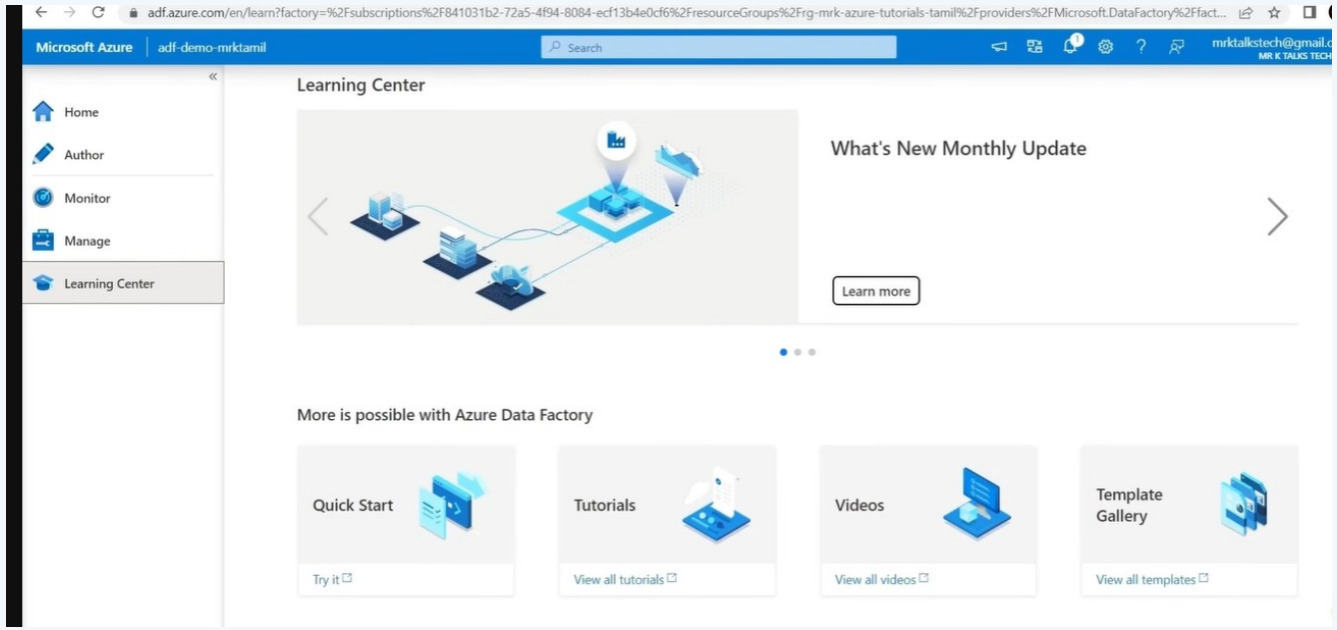
CMK provides more control over key access and rotation.

It requires correct key permissions and additional administration.

SCREENSHOT NOTE

No customer-managed key is configured in the screenshot.

Learning Center



PURPOSE

Use built-in learning resources to learn and practice Azure Data Factory.

KEY POINTS

Quick Start: introductory steps.

Tutorials: guided exercises.

Videos: visual demonstrations.

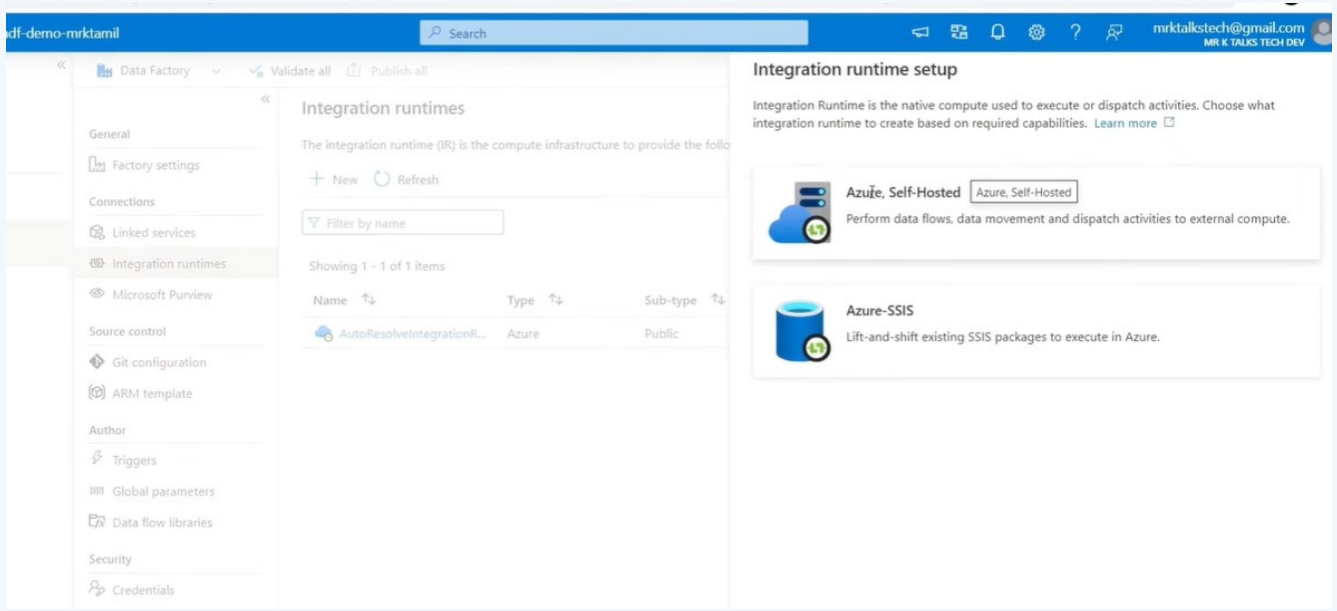
Template Gallery: starting points for common workflows.

Monthly updates: learn about new features.

USE CASE

Follow a tutorial to build a pipeline that copies data from Azure SQL Database to Data Lake Storage.

Create Integration Runtime



AZURE, SELF-HOSTED

Use a self-hosted runtime for connectivity to on-premises or private-network data sources and supported activities.

KEY POINTS

Install/configure the runtime on a server that can reach the source.

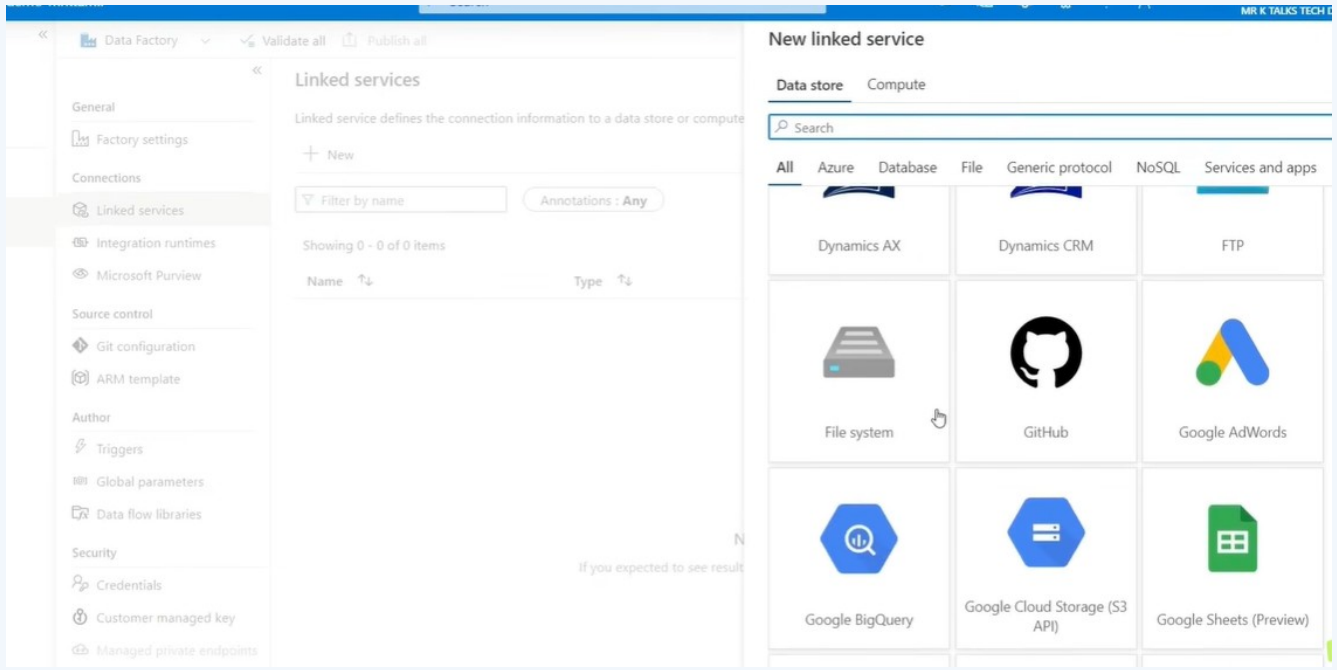
Azure-SSIS: provision runtime infrastructure to run supported SSIS packages in Azure.

Choose based on source location and workload.

USE CASE

Copy data from an on-premises SQL Server to Azure Data Lake Storage, or lift-and-shift existing SSIS packages to Azure.

New Linked Service – Connector Gallery



PURPOSE

Select a connector for the data store, application, or compute service you want to connect to.

KEY POINTS

Browse tabs such as Azure, Database, File, NoSQL, and Services and apps.

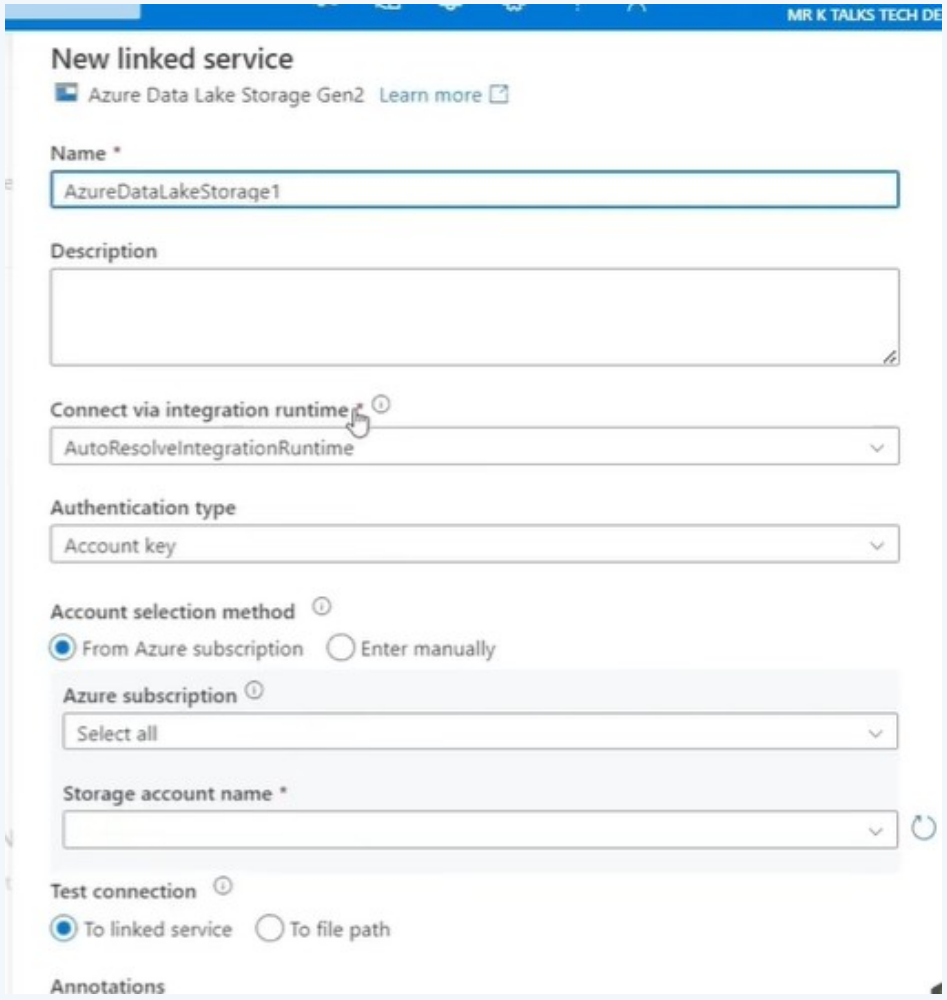
Examples shown: FTP, File system, GitHub, BigQuery, Google Cloud Storage, and Google Sheets (Preview).

Connector availability and authentication options vary by service.

USE CASE

Choose the source and destination connectors needed for a copy pipeline.

Linked Service – Azure Data Lake Storage Gen2



PURPOSE

Configure the connection from ADF to an Azure Data Lake Storage Gen2 account.

KEY POINTS

Name: identify the connection, e.g. AzureDataLakeStorage1.

Integration Runtime: select the runtime used for connectivity.

Authentication: Account key is selected in this screenshot.

Select the subscription/account or enter details manually.

Test connection to verify access.

USE CASE

Connect ADF to a storage account so a pipeline can read or write CSV, JSON, or Parquet files. For production, consider managed identity where supported.